Neighborhood-Scale Urban Heat Islands and Tree Canopy Inequity in Phoenix, Arizona

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***Abstract*— Phoenix, Arizona, is experiencing the effects of an extreme surface urban heat island (SUHI), which is exacerbating health risks in one of the hottest US metropolitan areas. This study examines the spatial variations of clear summer land surface temperature at neighborhood level and quantifies the cooling effects of vegetation cover compared to built-up areas. Using Google Earth Engine, we processed Landsat 8 and 9 Collection 2 level 2 images for the summer months of 2023 (June to September). We calculated the LST, the Normalized Difference Index of Variables (NDVI) and the Normalized Difference Build-up (NDBI) for 434 census tracts in the Phoenix metropolitan area. Analysis used zonal statistics, Pearson correlation, random forest regression and K-means clustering to determine the causes of thermal imbalance. The results show a strong negative correlation (r = -0.66) between the density of vegetation (NDVI) and LST, which confirms that the areas with a higher coverage of canopy are significantly colder. In contrast, the built-up intensity (NDBI) showed a strong positive correlation (r = 0.67) with the water surface temperature. Spatial clustering identified distinct hotspots (n=6) with an average LST of 60.2°C, compared to cold spots (n=40) with an average LST of 52.6°C, mainly due to vegetation differences. These findings provide a reproducible geospatial process to identify heat-sensitive communities and support the prioritization of green infrastructure in equitable urban planning.**

Keywords—Remote Sensing, Urban Heat Island, Land Surface Temperature, Environmental Equity, Google Earth Engine, Random Forest.

I. Introduction

A. Background and Motivation

Extreme heat is the leading cause of weather-related mortality worldwide, a risk that is particularly acute in urban settings. In Phoenix, Arizona, rapid urbanization and the replacement of natural soil by impervious surfaces have created a significant surface area heat index (SAHI). These heat islands are not uniformly distributed; rather, they are often correlated with socio-economic differences and variations in urban vegetation. As climate change increases the frequency and duration of heat waves, understanding the spatial distribution of heat is becoming a key issue for public health interventions.

B. Related Work

Previous research using satellite-derived land surface Remote sensing has become the standard for monitoring SUHI due to its ability to capture synoptic views of thermal patterns that sparse weather station networks cannot. Previous research using satellite-derived Land Surface Temperature (LST) has demonstrated the efficacy of these methods in mapping fine-scale heat vulnerability. For instance, Hulley et al. utilized ECOSTRESS and MODIS data to identify heat vulnerability in Los Angeles, highlighting the role of surface properties in diurnal heat retention. Similarly, recent microclimate simulations by Yang and LaRoche quantified the potential cooling effects of tree canopy specifically in the Phoenix region, and indicate that targeted greening can significantly reduce local temperature. However, the gap between high-resolution remote sensing data and administrative units (census tracts) used for policy making and planning still needs to be bridged.

C. Research Objectives

Building on this baseline work, the project aims to measure summer LST patterns at neighborhood level and to investigate their relationship with vegetation, hard surfaces and tree canopy distribution. In particular, the study addresses three fundamental research issues:

1) How does clear-sky summer LST vary among neighborhoods in the Phoenix metropolitan area?
2) To what extent do vegetation (NDVI) and built-up surfaces (NDBI) explain these thermal differences?
3) Can unsupervised machine learning effectively identify and profile the most vulnerable communities based on their thermal and landscape characteristics?

II. Study Area and Data

A. Study Area

The study area includes the incorporated boundaries of the metropolitan area of Phoenix, which is located in the Salt River valley. The area is characterized by a dry desert climate, with intense summer sunshine and low annual rainfall. The city is experiencing repeated extreme heat events, which makes it an ideal test bed for the SUHI analysis. The spatial extent was defined by a 20 km perimeter around the city center (Lat: 33.4484, Lon: -112.0740) to capture the urban-rural divide.

B. Data Sources

1) Satellite Imagery

We used Landsat 8 and 9 Collection 2 level 2 surface reflectivity and surface temperature products. These data sets provide heat infrared data (band 10) with a resolution of 100m (sampled at 30m) and optical bands with a resolution of 30m. The period of analysis was limited to the summer months (1 June to 30 September) of year 2023 in order to capture the peak heat conditions.

2) Census Data

Neighborhood boundaries have been defined using TIGER/Line Shapefiles (2020) shapefiles from the U.S. Census Bureau for Maricopa County, Arizona (State FP 04, County FP 013). This vector data file allows us to aggregate raster data at the pixel level into actionable administrative units.

3) *Elevation Data*

Topographic variables were derived from the USGS SRTM (Shuttle Radar Topography Mission) Digital Elevation Model (DEM) at 30m resolution. Elevation is a critical control variable in Phoenix due to the topographic relief of the surrounding valley.

III. Methodology

All geospatial processing was performed on Google Earth Engine (GEE), which leveraged its cloud computing capabilities to efficiently process a collection of multiple time-lapse images.

A. Image Preprocessing and Cloud Masking

The main problem in optical remote sensing is contamination of the atmosphere. We filtered the Landsat data for the study area and set a threshold for cloud coverage of less than 30 percent. While a strict cloud filter is ideal, the Arizona summer monsoon season requires a little tolerance to ensure that enough data is available. We used the bitwise cloud masking function to remove clouds and cloud shadows using the QA_PIXEL and QA_RADSAT zones. A median composite has been generated to produce a single, smooth, cloudless image of the 2023 summer, representing the average summer conditions.

B. Water Masking

To make sure that the statistics on temperature reflected human habitation and not natural water bodies, we used a water mask. A normalized difference in water intensity (NDWI) was calculated and pixels with a NDWI > 0 were eliminated from the analysis. This step is essential for accurate zoning statistics, as water bodies may artificially reduce the average temperature of the census area.

C. Derivation of Land Surface Temperature (LST)

Instead of using raw temperature data, we derived the land surface temperature using the standard emissivity correction process to take account of the diversity of surface materials in the urban environment. We used a Single Channel Algorithm, a generally accepted method for thermal analysis of Landsat. First, we calculate the Fraction of Vegetation Cover (FVC) using the dynamic NDVI range for distinguishing between bare land and thick vegetation. This FVC was then used to estimate the land surface emissivity, a critical variable that adjusts for the different materials (for example asphalt versus grass) that emit heat differently. Finally, the raw luminosity temperature from Landsat band 10 was converted into LST (in °C) by incorporating this emission factor, which ensured that the final temperature maps accurately reflected the physical characteristics of the urban surface and not just atmospheric noise.

D. Spectral Indices Calculation

We've calculated two other indices that serve as predictors of LST spatial patterns.

Normalized Difference Vegetation Index (NDVI): A proxy for green biomass and tree canopy density.

$$NDVI = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

Normalized Difference Built-up Index (NDBI): A proxy for impervious surface intensity (concrete, asphalt).

$$NDBI = \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}}$$

E. Zonal Statistics and Aggregation

To bridge the gap between the data in pixels and the units of politics, we have conducted a survey of zonal statistics. Mean values for LST, NDVI, NDBI and increase were aggregated for all census tracts (n = 434). Tracts with null data (due to masking or clipping) were filtered out to produce a clean dataset of 434 neighborhoods for statistical testing.

F. Statistical Modeling and Machine Learning

In analyzing the data, we used a multilateral statistical approach:

1) *Pearson Correlation:*

To quantify the linear relationship between temperature (LST) and surface area (NDVI, NDBI), we calculated the correlation coefficient (r) for all channels.

2) *Random Forest Regression:*

To evaluate nonlinear relationships and variable importance, we trained a random forest regressor on 100 trees. The model used LST as the target variable and predictors NDVI, NDBI and increase. To validate the model, we used the Out-of-Bag Error (OOB) estimation and extracted scores of importance to rank the heat drivers.

3) *K-Means Clustering:*

We used unsupervised K-means clustering to classify neighborhoods into three different profiles according to their biophysical signature (LST, NDVI, NDBI).

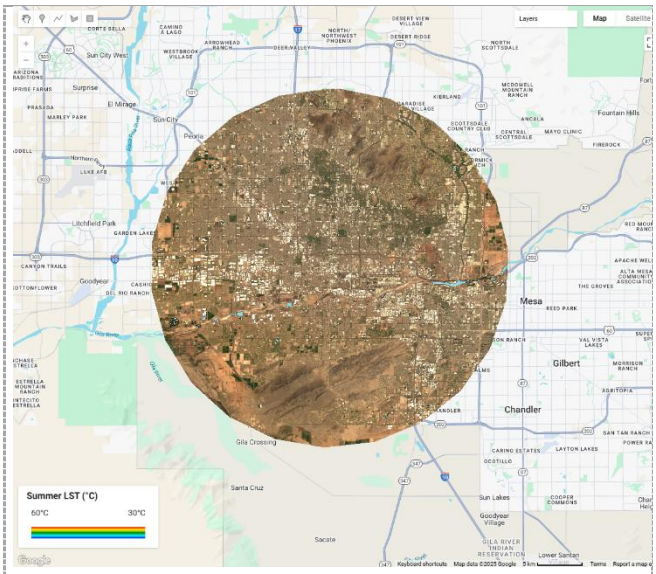
4) *Local Indicators of Spatial Association (LISA):*

We calculated Z-scores for each tract to identify statistically significant spatial outliers - neighborhoods that are abnormally hot (hot spots) or abnormally cold (cold spots) relative to the regional mean.

IV. Results

A. Spatial Distribution of Heat (LST)

LST map (Figure 1) shows a distinct thermal gradient in the center- peripheral area. The hottest temperatures are recorded in the urban core, which is characterized by dense commercial and industrial infrastructure, often reaching above 60°C. By contrast, well-established residential neighborhoods with mature landscaping and agricultural borders seem much colder. The LST spatial pattern reflects inversely the True Color image, where green vegetation is directly correlated with the lower surface temperature.



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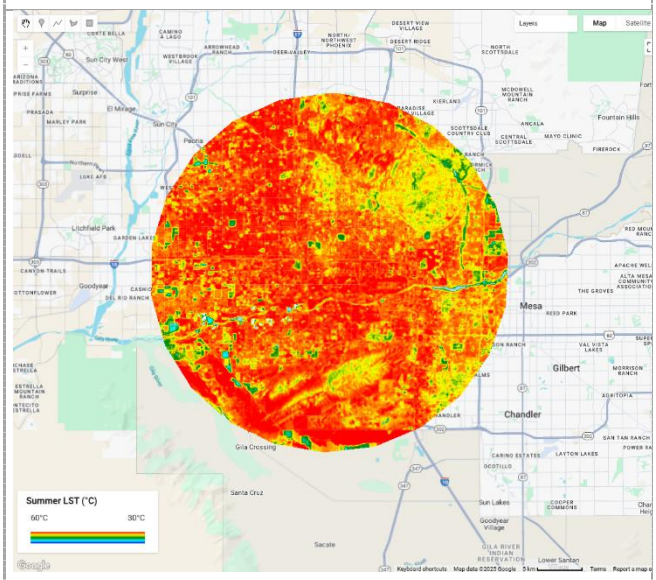
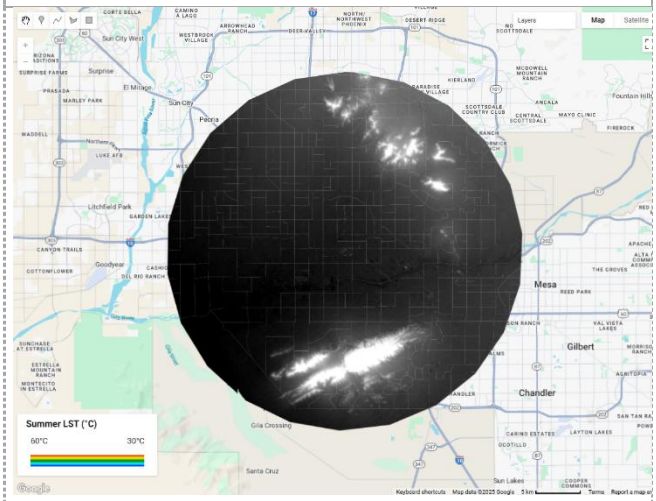
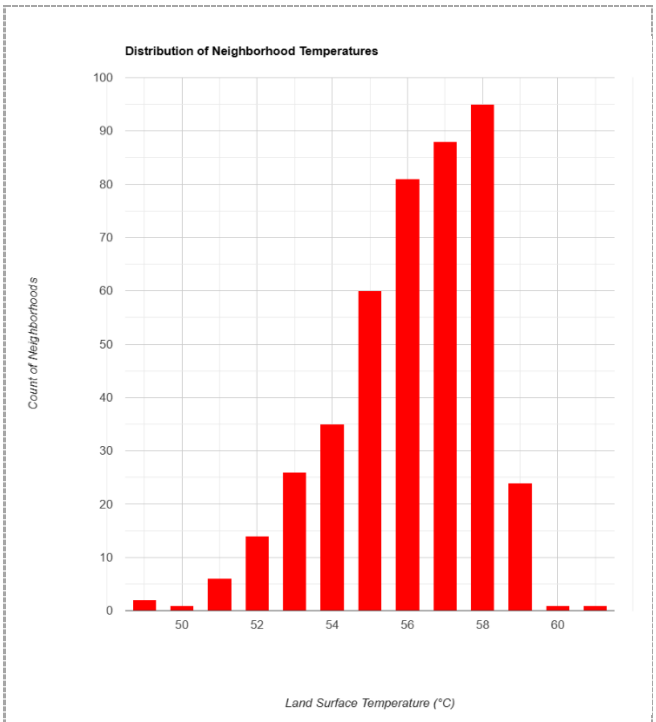


LST

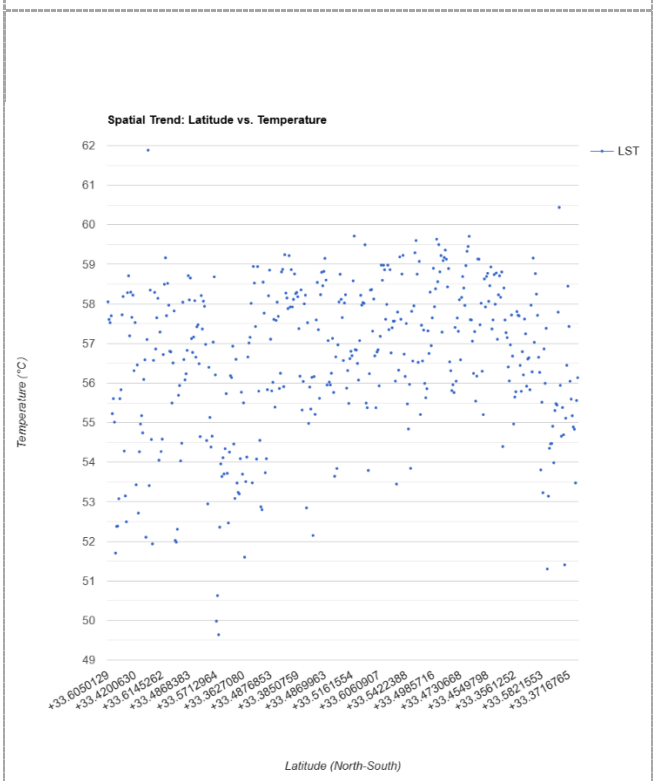


ELEVATION

Figure 1. Comparison of True Color imagery (upper) and derived Land Surface Temperature (middle), Elevation (lower) for Phoenix, Summer 2023



NEIGHBORHOOD TEMPERATURES



LATITUDE vs TEMPERATURE

Figure 2. Neighborhood Temperatures; Latitude vs Temperature

Beyond the visual gradient, the distribution of heat in quantitative terms is illustrated in Figure 2. The histogram shows a slightly skewed thermal profile with most neighbourhoods clustered between 55°C and 59°C, suggesting that extreme heat exposure is widespread rather than isolated to a few outliers. In addition, to test for general geographic determinants, a relationship between latitude and surface temperature was analysed. The scatter plot shows no significant linear north-south gradient, which confirms that temperature variation in Phoenix is driven by local surface heterogeneity, specifically soil cover and elevation, rather than by a wide latitude distribution.

B. Quantitative Analysis of Drivers

Statistical analysis confirms the hypothesis of the cooling effect. As can be seen in the scatter plot (Figure 1) In the table I, there is a strong linear relationship between the surface area and temperature.

1) Vegetation: NDVI and LST showed a strong negative correlation (r = -0.66). This suggests that as the density of vegetation increases, the surface temperature decreases linearly..

2) Impervious Surfaces: NDBI showed a strong positive correlation (r = 0.67) with LST, which confirmed that concrete and asphalt are the main heat sinks on the surface.

3) Inter-variable Relationship: The strong negative correlation between NDVI and NDBI (r = -0.77) underlines the trade-off between land use in the city: concrete-dedicated land is land taken out of vegetation for cooling.

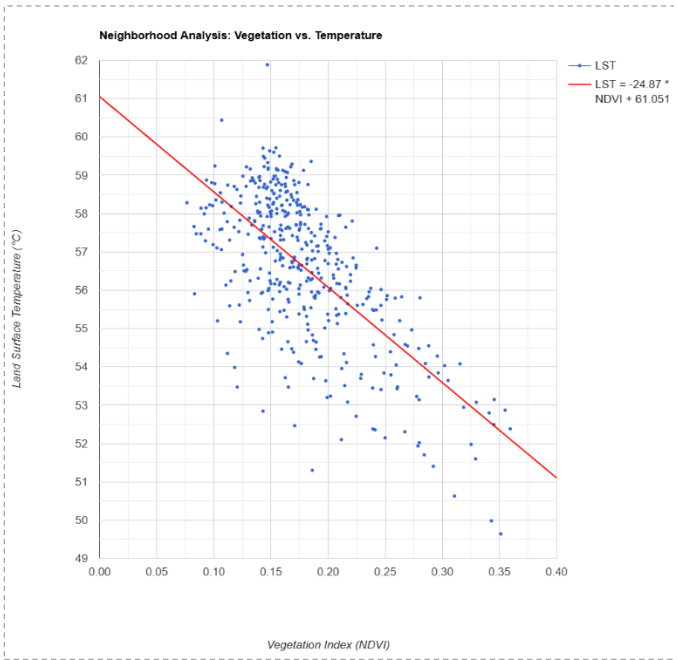


Figure 3. Scatter plot illustrating the negative correlation between Vegetation Index (NDVI) and Land Surface Temperature (LST) across 434 census tracts.

Variable Pair	Pearson Correlation (r)	Interpretation
LST vs. Vegetation (NDVI)	-0.66	Strong Negative (More trees = Lower Temp)
LST vs. Built-Up (NDBI)	0.67	Strong Positive (More concrete = Higher Temp)
NDVI vs. NDBI	-0.77	Strong Negative (Vegetation and Built-up are inverse)

Table 1. Correlation Analysis

C. Machine Learning Insights (Random Forest)

While correlation suggests linear relationships, the random forest model provides a more detailed view of variable importance (Figure 3). Interestingly, Elevation was the top predictor, together with NDVI. This reflects the topography of the Phoenix valley, where small changes in elevation can affect air flow and exposure. However, among the variables that can change in an urban context, the NDVI remained the dominant predictor, far outperforming the NDBI. This suggests that the addition of green infrastructure (trees) can be more effective in mitigating the heat than simply lowering the density of buildings.

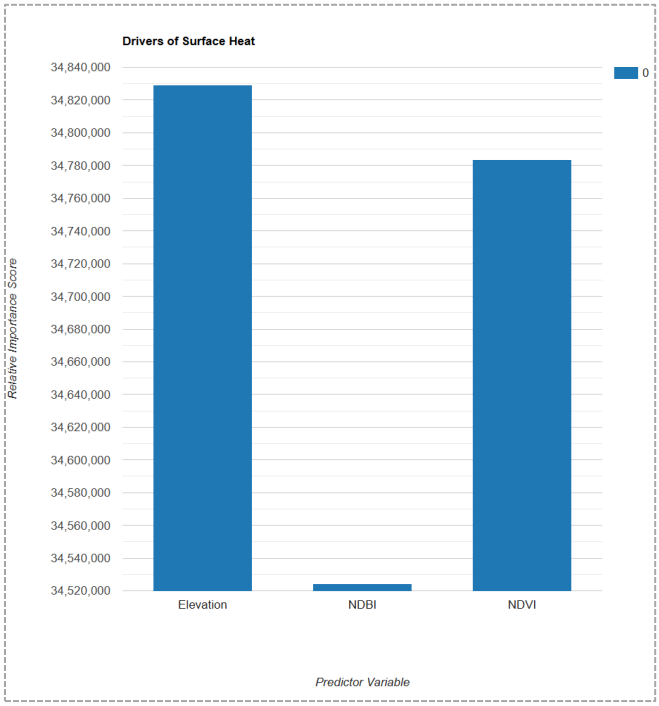


Figure 4. Random Forest Variable Importance scores. Elevation and NDVI are the primary drivers of LST variation, with NDBI playing a secondary role.

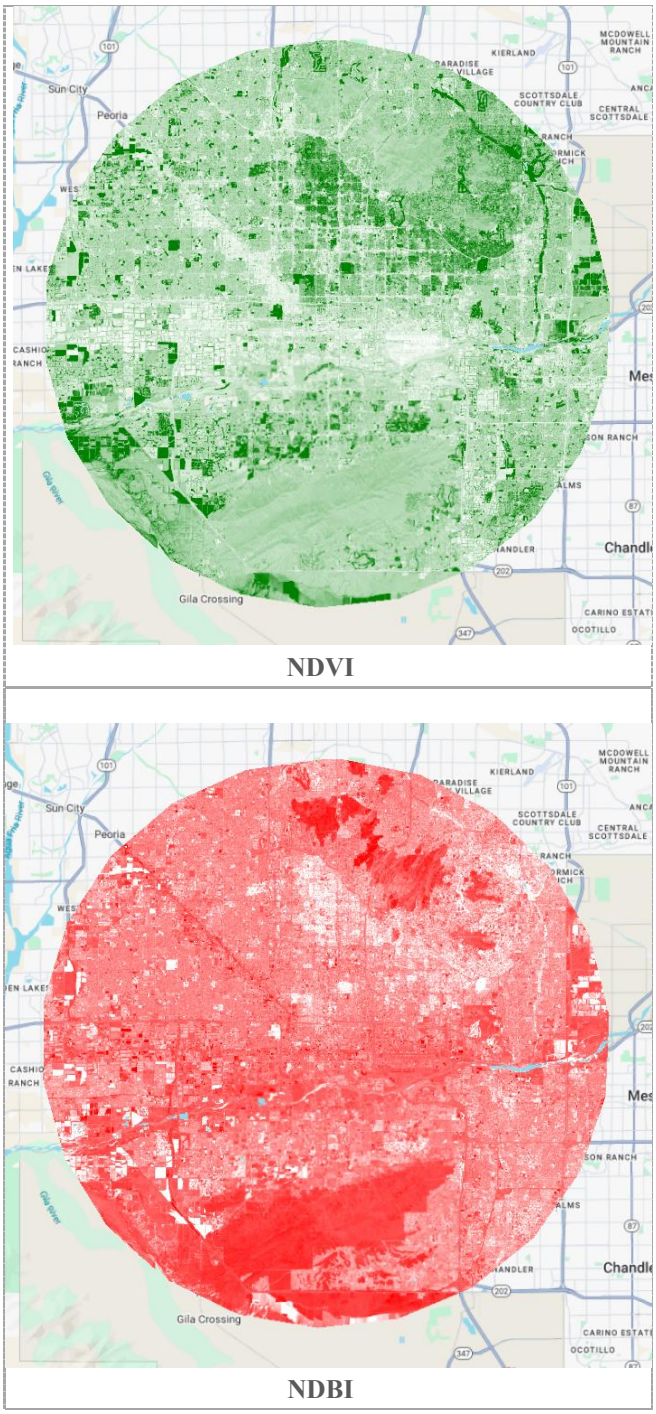


Figure 5. NDVI vs NDBI

Spatial vegetation patterns (Figure 4b, NDVI) and impermeable surfaces (Figure 4c, NDBI) show a stark contrast. The dense clustering of high NDBI values in the urban core is spatially aligned with hotspots of heat, while the broken green corridors in figure 4b provide the only significant cooling counterforce.

D. Thermal Inequity and Clustering

K-means clustering and LISA (Z-score) analysis successfully segmented the city into distinct thermal environments. We identified 6 critical hot spot and 40 critical cold spot tracts (Figure 6).

- The Thermal Gap: The average difference in temperature between hot spots (60.2 °C) and cold spots (52.6 °C) is 7.6 °C.
- The Vegetation Gap: Hot spots have a mean NDVI of only 0.142, while cool spots have an NDVI of 0.262 - almost twice as high as the density of vegetation (Table 2).

Cluster Type	Count of Neighbourhoods	Mean Temp (LST °C)	Mean Vegetation (NDVI)	Mean Built-Up (NDBI)
Cool Spots	40	52.6°C	0.262	-0.072
Neutral	388	57.0°C	0.171	-0.019
Hot Spots	6	60.2°C	0.142	0.015

Table 2. Characteristics of Thermal Clusters

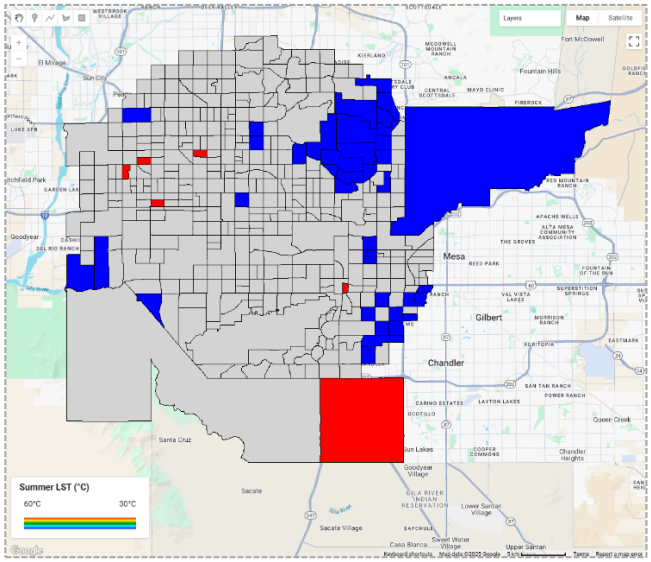


Figure 6. Spatial clustering of LST Anomalies (Z-Scores) highlighting critical Hot Spots (Red) and Cool Spots (Blue)

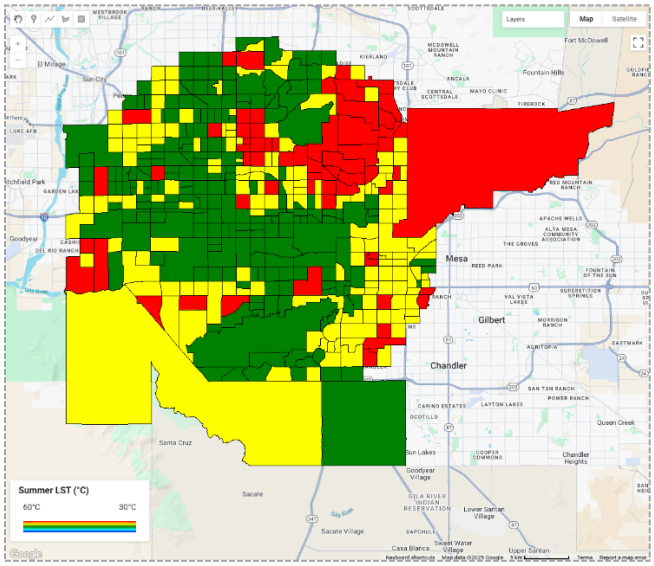


Figure 7. K-Means Clustering Results showing the three distinct neighborhood profile based on biophysical similarity.

E. Temporal Trends (2018–2023)

To contextualize the 2023 data, we have analyzed five-year summer temperature series. Trend line (Figure 8) shows a slight but steady upward trend in raw surface temperatures. This suggests that despite awareness of the heat island effect, the region is not cooling; rather, the heat load is increasing slowly, which reinforces the urgency of the findings of the study.

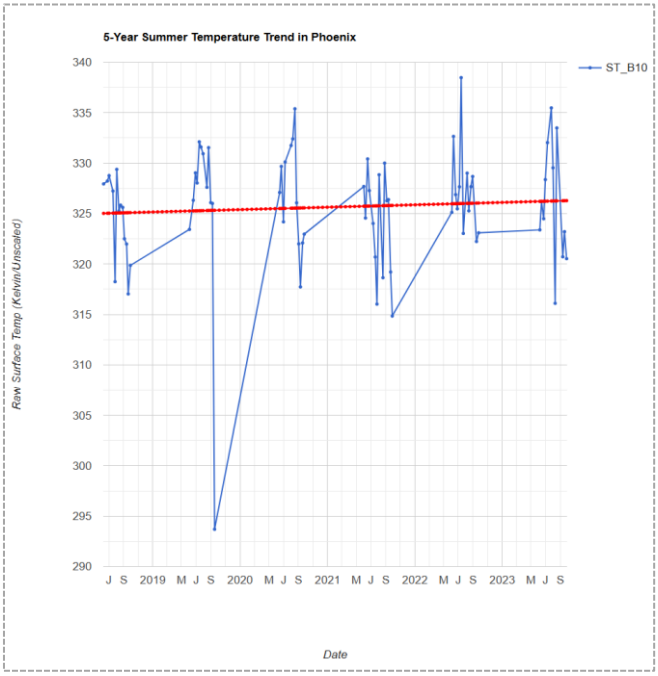


Figure 8. Five-year trend of summer surface temperatures (2018–2023) derived from Landsat thermal bands.

V. Thermal Inequity and Clustering

A. The Role of Green Infrastructure

The results strongly support the hypothesis that vegetation is a primary variable in the Phoenix neighborhood microclimate. The 7.6°C difference between the most and least vegetated neighbourhoods represents a significant disparity in exposure to environmental health. This is consistent with the findings of Yang and La Roche, who demonstrated by simulation that tree canopy is essential for cooling. Our satellite observations confirm that this cooling effect is measurable and significant in the vicinity.

B. Urban Planning Implications

The strong inverse correlation between NDVI and NDBI indicates that there is competition for the space. In many Hot Spot neighborhoods, the NDBI tallies show that the land is almost entirely paved or built on, with little room for tree-planting. This structural constraint means that traditional tree planting may not be sufficient; green roofs, vertical gardens or the removal of underused parking spaces may be needed to create permeable and plantable areas.

C. The Elevation Factor

Random Forest analysis shows that the elevation is a major temperature regulator. This poses a challenge to city planners: they cannot change the topography of the city. Thus, low-lying areas are naturally prone to heat build-up. These areas should be the main beneficiaries of public funding for heat mitigation (e.g. shade structures, cooling centers), as their geography places them at a natural disadvantage which cannot be fully compensated by vegetation alone.

D. Limitations

This study is based on LST from satellites, which measure the surface temperature of the earth, not ambient air temperatures that people feel. Although LST and air temperatures are correlated, LST is often much higher (for example asphalt may have a temperature of 60°C, while air is 40°C). Moreover, the analysis was limited to days with clear skies. The summer monsoons bring clouds that block the satellites, so our data represent the hottest and brightest days of the year, which can skew results to extremes.

VI. Conclusion and Future Directions

This project has successfully developed a reproducible geospatial workflow for identifying surface urban heat island (SUHI) hotspots at neighborhood scale, using Google Earth Engine. We have shown that the main statistical cause of the extreme heat is the lack of vegetation in the most vulnerable census tracts in Phoenix. The identification of a 7.6°C temperature difference between neighborhoods highlights the environmental inequity inherent in the current urban form of the city.

In order to verify the “inequity", future analyses will include ACS data from the US Census. As LST is already aggregated by census tract, socio-economic variables can be linked to GEOID1 by way of a link to the GEOID database. Specifically, median household income will allow the `luxury effect' to be tested, while data on race and ethnicity will explicitly test whether the 7.6°C thermal gap disproportionately affects minority communities. In addition, age structure data will help to identify areas where there is an overlap in temperature exposure with vulnerable older populations.

While the current Random Forest model has identified global heat drivers such as elevation and vegetation, future work should use Geographic Weighted Regression (GWR). Global models assume stable relationships, but GWR can detect spatial discontinuity. This approach will reveal whether the cooling effect of the tree canopy (r=-0.66) varies at the local level and whether green infrastructure provides a higher critical cooling value in dense urban areas than in more suburban areas.

The goal is to synthesize the findings into an actionable heat index for urban planners. By normalizing and combining the physical exposure (LST) and layers of social vulnerability (income, age) the analysis may produce a composite risk score. This will refine the current grouping by identifying high-risk zones where extreme heat is combined with low adaptive capacity, and offer a more nuanced tool for prioritizing public funding than temperature data alone.

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